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Centre Number

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Student Number

2021 YEAR 12

Advanced Mathematics

Task 2

Monday 1st March, 2021

Q	Marks
MC	
5	
6	
7	
8	
9	
10	
11	
12	
Total	

General Instructions:

- Reading time – 5 minutes
- Working time – 55 minutes
- Write using blue or black pen
- NESA approved calculators may be used
- Show relevant mathematical reasoning and/or calculations

Total Marks:
34

Section I - 4 marks

- Allow about 7 minutes for this section

Section II - 30 marks

- Allow about 48 minutes for this section

This question paper must not be removed from the examination room.

This assessment task constitutes 20% of the course.

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Section I

4 marks

Allow about 7 minutes for this section

Use the multiple-choice sheet for Question 1–4

1 Find the value of $\log_4 93$, correct to 3 significant figures.

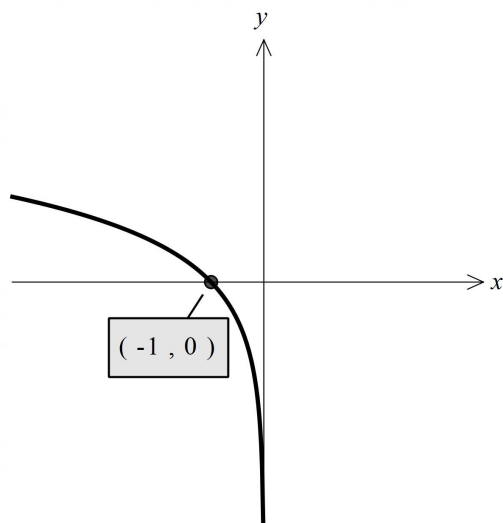
- (A) 1.97
- (B) 1.968
- (C) 3.27
- (D) 3.270

2 Find $\frac{d}{dx} \cos(x + 1)$

- (A) $\cos(x + 1)$
- (B) $\sin(x + 1)$
- (C) $-\sin(x + 1)$
- (D) $-\cos x \sin(x + 1)$

Examination continues on next page

- 3 Differentiate $e^x \sin x$ with respect to x .
- (A) $e^x \cos x$
- (B) $-e^x \cos x$
- (C) $e^x(\sin x + \cos x)$
- (D) $e^x(\sin x - \cos x)$
- 4 Which equation corresponds to the graph below?



- (A) $y = -\ln(x)$
- (B) $y = \ln(-x)$
- (C) $y = e^{-x} - 1$
- (D) $y = -e^{-x}$

End of Section I

Section II

30 marks

Allow about 48 minutes for this section

In Questions 5 – 12, your response should include relevant mathematical reasoning and/or calculations.

Question 5 (5 marks)

Differentiate the following, with respect to x , and fully simplify:

(a) $f(x) = 6x - 7 + e^{-2x}$ 1

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(b) $y = 3^{-x}$ 1

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(c) $y = \log_e(\sec x)$ 3

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Examination continues on next page

Question 6 (4 marks)

Consider the stem and leaf plot below with 17 scores.

0	58
1	04668999
2	00026
3	00

- (a) Give the 5-point summary.

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- (b) State the value of any outliers. Justify your answer with calculations.

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Question 7 (4 marks)

Consider the function $f(x) = 3 - e^{-x}$

- (a) Evaluate $\lim_{x \rightarrow \infty} f(x)$

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- (b) Sketch the graph of $f(x)$, clearly stating the exact value of all intercepts.

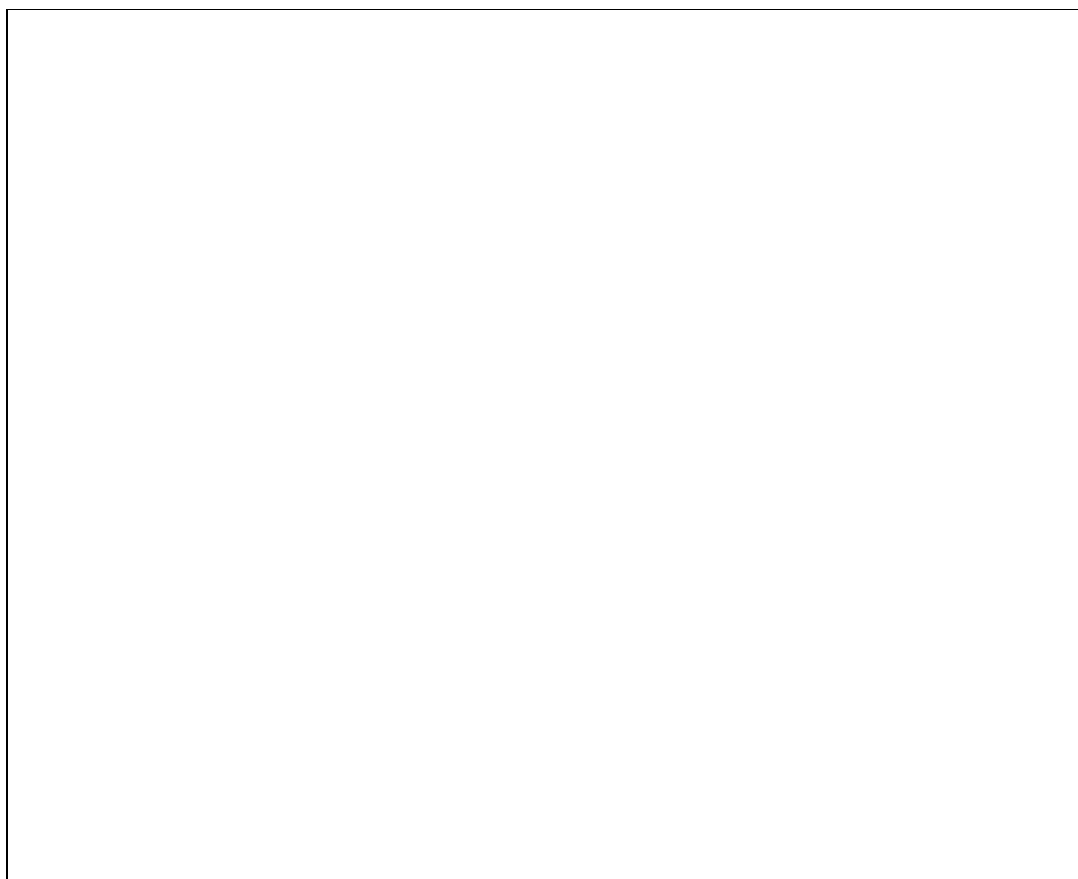
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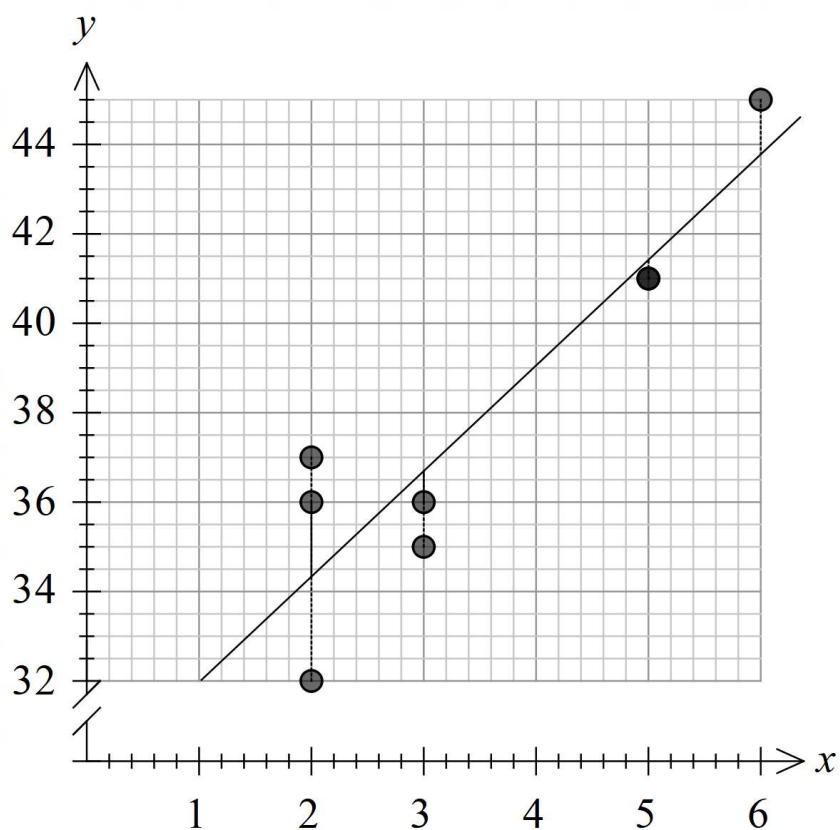
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Question 8 (4 marks)

A set of data was collected comparing the age of cars to their stopping distance during a brake test.

The collected data is shown in the table and graphed in the scatterplot shown.

Age of car (years)	2	2	5	3	6	2	5	3
Stopping Distance (m)	36	37	41	35	45	32	42	36



- (a) Find the Pearson correlation coefficient, r , correct to 3 decimal places.

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- (b) Find the equation of the least-squares line of best fit, giving the coefficients to 1 decimal place. **1**

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- (c) Use your result from part (b) to predict the stopping distance of car that is 4 years old. Comment on the reliability of your prediction. **2**

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Examination continues on next page

Question 9 (2 marks)

If $\log_{10} x = 0.6$ and $\log_{10} y = 0.2$, evaluate $\log_{10} \left(\frac{x}{\sqrt{y}} \right)$.

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Question 10 (2 marks)

Given that $f(x) = \tan^3 x$, calculate $f' \left(\frac{\pi}{4} \right)$.

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Question 11 (4 marks)

Super Mine World 64 is a video game known for "speedrunning", where players aim to complete the game in as short a run time as possible. A list of the world record fastest times are listed in the table below to the nearest second.

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840s	848s	859s	912s	924s	931s	937s	939s	951s
------	------	------	------	------	------	------	------	------

When a new score was added to this list, the sample mean and interquartile range both decreased.

Using interval notation, state the interval in which this new score must lie.

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Question 12 (5 marks)

The energy, E Megajoules, radiated by an earthquake is given by the formula

$$E = k \times 10^{1.5R}$$

where R is the magnitude of the earthquake on the Richter scale and k is a constant. An earthquake of magnitude 3 radiates 2000 Megajoules of energy.

- (a) Show that $\log_{10} E = \log_{10} 2 + 1.5(R - 1)$ **3**

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- (b) Find the magnitude of an earthquake which radiates 5 million Megajoules of energy correct to 2 decimal places. **2**

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End of examination

Extra writing space

If you use this space, clearly indicate which question you are answering.

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Section I

4 marks

Allow about 7 minutes for this section

Use the multiple-choice sheet for Question 1–4

1 Find the value of $\log_4 93$, correct to 3 significant figures.

(A) 1.97

(B) 1.968

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2 Find $\frac{d}{dx} \cos(x + 1)$

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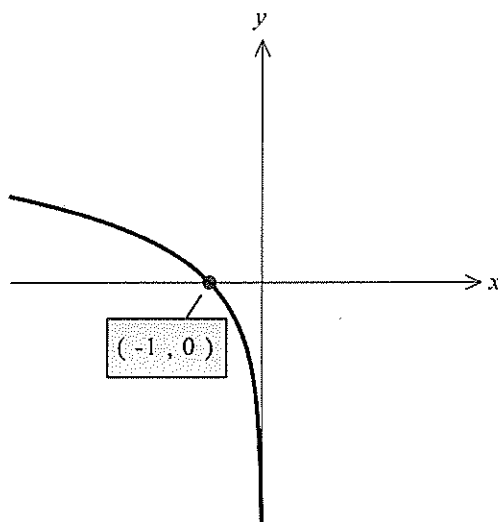
(A) $e^x \cos x$

(B) $-e^x \cos x$

(C) $e^x(\sin x + \cos x)$

(D) $e^x(\sin x - \cos x)$

4 Which equation corresponds to the graph below?



(A) $y = -\ln(x)$

(B) $y = \ln(-x)$

(C) $y = e^{-x} - 1$

(D) $y = -e^{-x}$

End of Section I

Section II

30 marks

Allow about 48 minutes for this section

In Questions 5 – 12, your response should include relevant mathematical reasoning and/or calculations.

Question 5 (5 marks)

Differentiate the following, with respect to x , and fully simplify:

(a) $f(x) = 6x - 7 + e^{-2x}$ 1

$$f'(x) = 6 - 2e^{-2x}$$

(b) $y = 3^{-x}$ 1

$$y' = -\ln 3 \times 3^{-x}$$

(c) $y = \log_e(\sec x)$ 3

Let $u = \sec x$

$$= (\cos x)^{-1}$$

$$u' = -(-\sin x)(\cos x)^{-2}$$

$$= \frac{\sin x}{\cos^2 x}$$

1 mark for derivative
of $\sec x$

$$y = \log_e u$$

$$y' = \frac{u'}{u}$$

$$= \frac{\frac{\sin x}{\cos^2 x} \times \cos x}{\frac{1}{\cos x}}$$

$$= \frac{\sin x}{\cos x}$$

$$= \tan x$$

1 mark for
simplifying

1 mark for correctly
applying

$$\frac{d \ln f(x)}{dx} = \frac{f'(x)}{f(x)}$$

Examination continues on next page

Question 6 (4 marks)

Consider the stem and leaf plot below with 17 scores.

0	58	Median
1	04668999	
2	00026	
3	00	

- (a) Give the 5-point summary.

$$\frac{n+1}{2} = \frac{17+1}{2} = 9^{\text{th}} \text{ score}$$

2

$$x_{\text{m.h.}} = 5$$

$$Q_1 = 15$$

$$Q_2 = 19$$

$$Q_3 = 21$$

$$Q_4 = 30$$

-1 mark for any mistakes

- (b) State the value of any outliers. Justify your answer with calculations.

2

$$IQR = 21 - 15 = 6$$

$\therefore 5$ is an outlier as $5 < 6$

$$1.5 \times IQR = 9$$

$$\text{Upper fence} = 21 + 9 = 30$$

$$\text{Lower fence} = 15 - 9 = 6$$

1 mark for IQR

1 mark for stating outlier is less than the lower fence = 6

Do NOT write in this area.

Question 7 (4 marks)

Consider the function $f(x) = 3 - e^{-x}$

- (a) Evaluate $\lim_{x \rightarrow \infty} f(x)$

1

$$= 3 - 0$$

$$= 3$$

- (b) Sketch the graph of $f(x)$, clearly stating the exact value of all intercepts.

3

When $x = 0$

$$f(0) = 3 - e^0$$

$$= 3 - 1$$

$$= 2$$

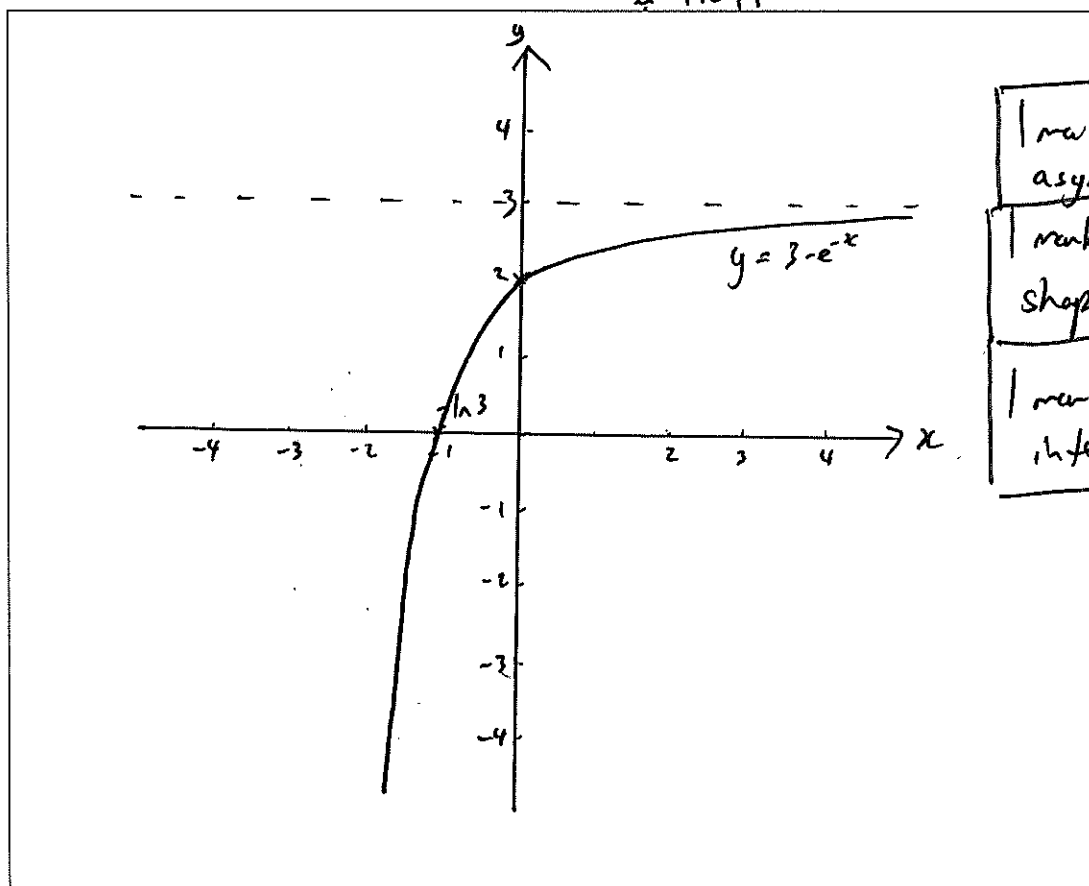
When $f(x) = 0$

$$3 - e^{-x} = 0$$

$$3 = e^{-x}$$

$$\ln 3 = -x$$

$$x = -\ln 3 \text{ or } x = \ln \frac{1}{3}$$



1 mark for asymptote

1 mark for shape

1 mark for both correct intercepts

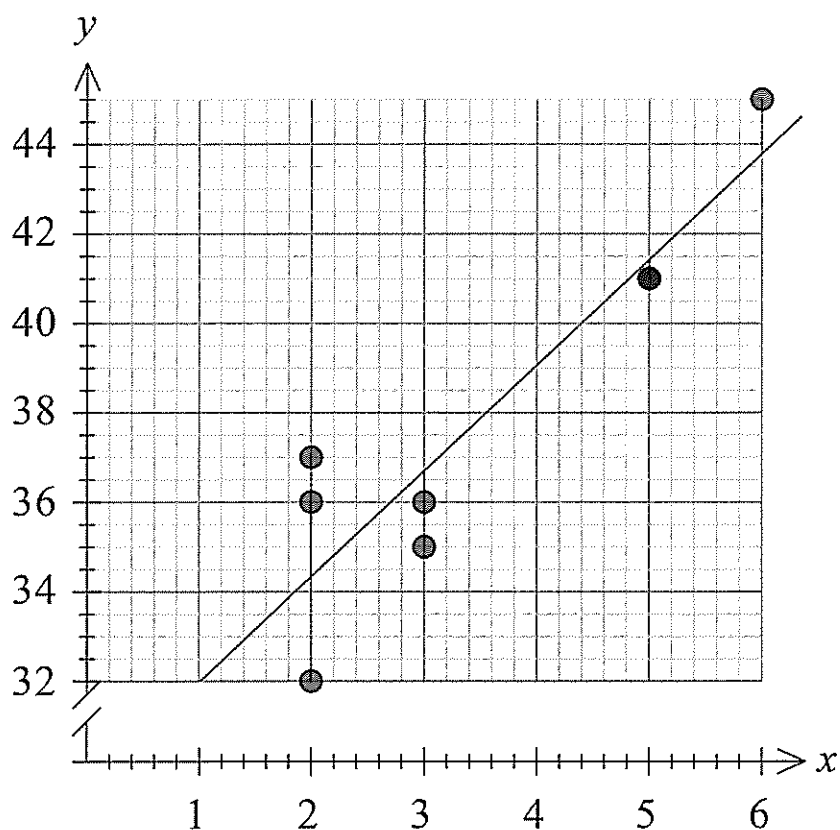
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The collected data is shown in the table and graphed in the scatterplot shown.

Age of car (years)	2	2	5	3	6	2	5	3
Stopping Distance (m)	36	37	41	35	45	32	42	36



- (a) Find the Pearson correlation coefficient, r , correct to 3 decimal places.

1

$r = 0.917$
Using calculator

-1 mark for incorrect rounding

- (b) Find the equation of the least-squares line of best fit, giving the coefficients to 1 decimal place. 1

$$A = 29.4$$

$$y = 29.4 + 2.4x$$

$$B = 2.4$$

or

Using calculator

Stopping distance = $29.4 + 2.4 \times \text{Age of car}$

- (c) Use your result from part (b) to predict the stopping distance of car that is 4 years old. Comment on the reliability of your prediction. 2

$$A + 4B = 39.2 \text{ (1 d.p.)}$$

Using calculator

1 mark (must not have used rounded values from (b))

Prediction is reliable as it is interpolating using the trend.

1 mark for reasoning "reliable" and "interpolation"

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Examination continues on next page

Question 9 (2 marks)

If $\log_{10} x = 0.6$ and $\log_{10} y = 0.2$, evaluate $\log_{10} \left(\frac{x}{\sqrt{y}} \right)$.

2

$$\begin{aligned} \log_{10} \left(\frac{x}{\sqrt{y}} \right) &= \log_{10} x - \log_{10} \sqrt{y} \\ &= \log_{10} x - \log_{10} y^{\frac{1}{2}} \\ &= \log_{10} x - \frac{1}{2} \log_{10} y \\ &= 0.6 - \frac{1}{2} \times 0.2 \\ &= 0.5 \end{aligned}$$

1 mark for separating two logs

1 mark for final result

Question 10 (2 marks)

Given that $f(x) = \tan^3 x$, calculate $f' \left(\frac{\pi}{4} \right)$.

2

$$f'(x) = 3 \tan^2 x \times \sec^2 x$$

1 mark for derivative

$$\begin{aligned} f' \left(\frac{\pi}{4} \right) &= 3 \times 1^2 \times (\sqrt{2})^2 \\ &= 6 \end{aligned}$$

1 mark for final result

Do NOT write in this area.

Question 11 (4 marks)

Super Mine World 64 is a video game known for "speedrunning", where players aim to complete the game in as short a run time as possible. A list of the world record fastest times are listed in the table below to the nearest second.

4

840s	848s	859s	912s	924s	931s	937s	939s	951s
------	------	------	------	------	------	------	------	------

When a new score was added to this list, the sample mean and interquartile range both decreased.

Using interval notation, state the interval in which this new score must lie.

$$\bar{x} = 904.5$$

$$Q_1 = \frac{848 + 859}{2}$$

$$= 853.5$$

$$Q_3 = \frac{937 + 939}{2}$$

$$= 938$$

$$IQR = 84.5$$

Let y be the new score.

As $y \leq 904$ and the list is now 10 scores,

the new Q_3 becomes 937

$$937 - 84.5 = 852.5$$

$$\therefore 852.5 \leq y \leq 904.5$$

However, the list is quoted to the nearest second.

1 mark for recognising new $Q_3 = 937$

1 mark for $Q_3 - IQR$

1 mark for $\bar{x}$ and IQR

$$\therefore 853 \leq y \leq 904$$

or

$$[853, 904]$$

1 mark for either interval notation and rounded integers

Examination continues on next page

Question 12 (5 marks)

The energy, E Megajoules, radiated by an earthquake is given by the formula

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where R is the magnitude of the earthquake on the Richter scale and k is a constant. An earthquake of magnitude 3 radiates 2000 Megajoules of energy.

- (a) Show that $\log_{10} E = \log_{10} 2 + 1.5(R - 1)$

3

$$\begin{aligned} 2000 &= k \times 10^{1.5 \times 3} \\ k &= \frac{2000}{10^{4.5}} \quad \boxed{\text{1 mark for finding } k \text{ in any form}} \\ &= 2 \times 10^3 \times 10^{-4.5} \\ &= 2 \times 10^{-1.5} \\ E &= 2 \times 10^{-1.5} \times 10^{1.5R} \\ &= 2 \times 10^{1.5R - 1.5} \\ &= 2 \times 10^{1.5(R-1)} \\ \log_{10} E &= \log_{10} (2 \times 10^{1.5(R-1)}) \quad \boxed{\text{1 mark for correctly taking } \log_{10} \text{ of both sides}} \\ &= \log_{10} 2 + \log_{10} 10^{1.5(R-1)} \\ &= \log_{10} 2 + 1.5(R-1) \quad \boxed{\text{1 mark for final result with correct working}} \end{aligned}$$

- (b) Find the magnitude of an earthquake which radiates 5 million Megajoules of energy correct to 2 decimal places.

2

$$\begin{aligned} \log_{10} E &= \log_{10} 2 + 1.5(R-1) \\ \log_{10} E - \log_{10} 2 &= 1.5(R-1) \\ \log_{10} \left(\frac{E}{2} \right) &= 1.5(R-1) \\ R-1 &= \frac{\log_{10} \left(\frac{E}{2} \right)}{1.5} \\ R &= \frac{\log_{10} \left(\frac{E}{2} \right)}{1.5} + 1 \end{aligned}$$

When $E = 5000000$

$$\begin{aligned} R &= \frac{\log_{10} \left(\frac{5000000}{2} \right)}{1.5} + 1 \\ &= 5.27 \text{ (2 d.p.)} \end{aligned}$$

$\boxed{\text{1 mark for substitution}}$
 $\boxed{\text{1 mark for correct final result.}}$

End of examination